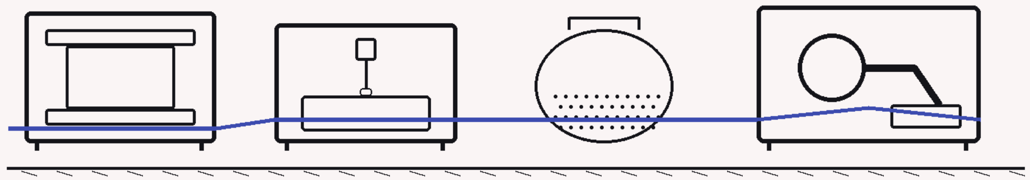
Five material classes could unlock 5–12 $GtCO_2$/year – if the predictions sent to laboratories are corrected and verified first.

Cobalt-free cathodes. Halide solid electrolytes. Direct-air-capture sorbents. Ammonia catalysts. Lead-free perovskites. The opportunity is climate-scale; the bottleneck is whether simulation results are reliable enough to spend scarce laboratory time on.



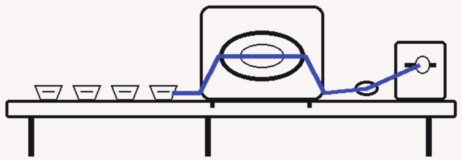
Lupine Science, “Five Materials That Could Unlock 5–12 $GtCO_2$/Year,” lines 2–16.

380,000 computationally stable structures; 736 independently synthesized by late 2023: a reported 0.2% validation rate.

380,000 **736**

REPORTED 0.2% VALIDATION RATE

Generation now outruns validation. At climate scale, more candidates do not help if errors survive generation, prediction, synthesis, and validation.



Validation-rate indicator; not a universal synthesizability probability.

Lupine Science, “The 0.2% Synthesis Problem,” lines 7–16; Merchant et al., Nature 624 (2023), DOI 10.1038/s41586-023-06735-9.

Foundation MLIPs were measured across 21 materials, up to 9 properties, and 228 published reference values.

21

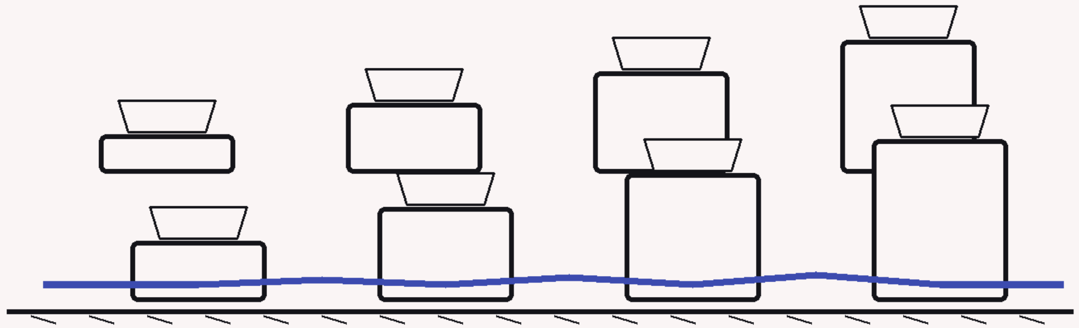
MATERIALS

up to 9

PROPERTIES

228

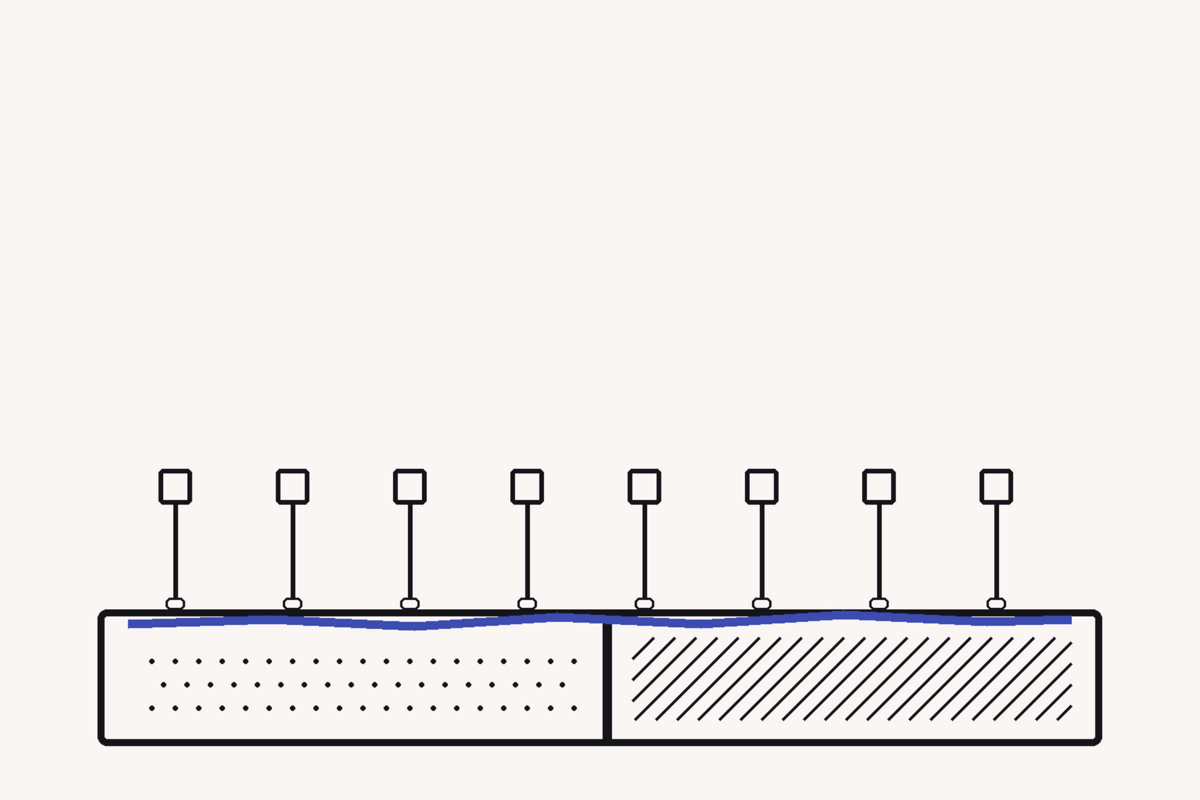
REFERENCE VALUES



In the measured benchmark, rankings often survive even when hard-property magnitudes miss by tens of percent. Bounded to the measured benchmark domain.

Lupine Science, “The Order Is Right, the Size Is Wrong,” lines 5–18 and 41–63.

A measured physical error field runs beside the live MLIP as an additive energy term with analytic forces. No fine-tuning. No retraining.



EXISTING MLIP

↓ MEASURED ROUTE

MEASURED ERROR FIELD

↓ ADDITIVE TERM

CORRECTED ENERGY + ANALYTIC FORCES

NO FINE-TUNING / NO RETRAINING

Lupine Science, “Investing in the Trust Layer,” lines 51–63; “The Order Is Right, the Size Is Wrong,” lines 67–81.

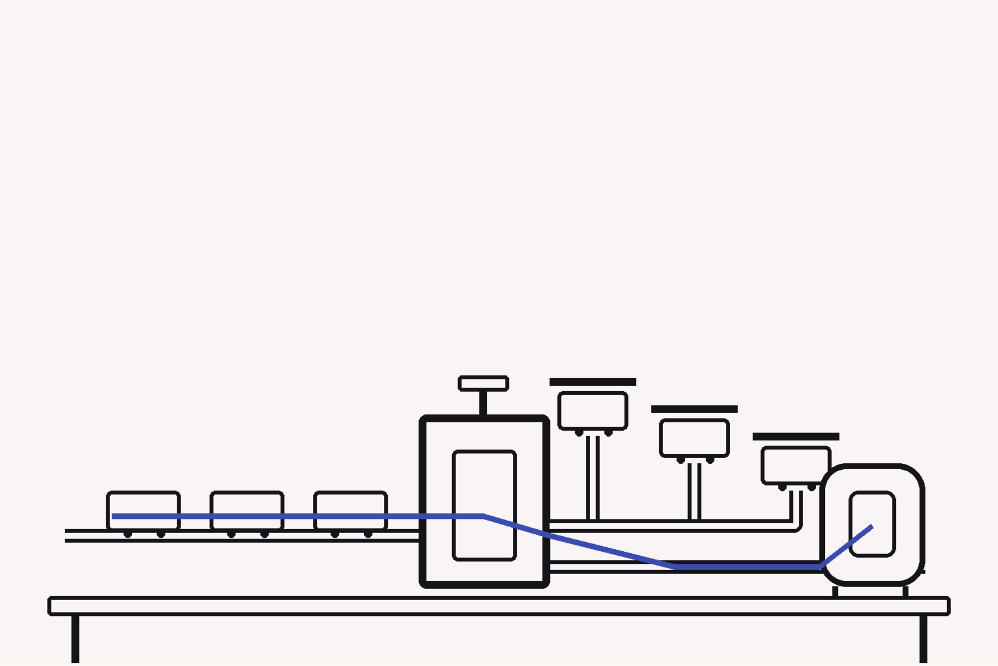
Every prediction takes one of three routes: measured correction, bounded claim, or abstention when evidence is insufficient.

MEASURED CORRECTION

BOUNDED CLAIM

ABSTAIN

The proof layer licenses corrections only where evidence supports them. The honesty proof is preserved in public: the first correction made alloy predictions worse; the resulting Lean law now blocks wrong-direction correction, and unsupported cases abstain.



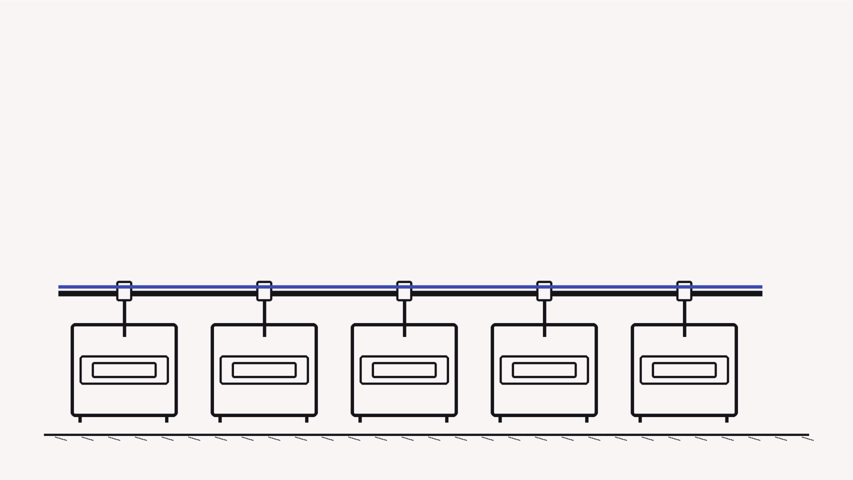
The first correction made predictions worse; the Lean law blocks wrong-direction correction.

Lupine Science, “The Correction That Hurt, and the Theorem That Stopped It,” lines 19–35 and 49–55.

$14.65 per 129 anchors. A completed, source-locked execution record.

$14.65 per 129 anchors

APPROVED PUBLIC COST BASIS



The evidence program spans **42 published Li-ion electrolyte chemistries**: a **30-path frozen test panel** plus **12 disjoint training chemistries**, evaluated with guidance from **4 foundation MLIPs**.

Two guidance models found the true extrema on every guided path; two recorded a 6.8 meV undercoverage deficit. FOR EDITOR REVIEW; not peer-reviewed.

Lupine Science, “The Union Verdict,” lines 17–35 (FOR EDITOR REVIEW; not peer-reviewed); “The Z1 Barrier Panel,” lines 1–17 and 23–66.

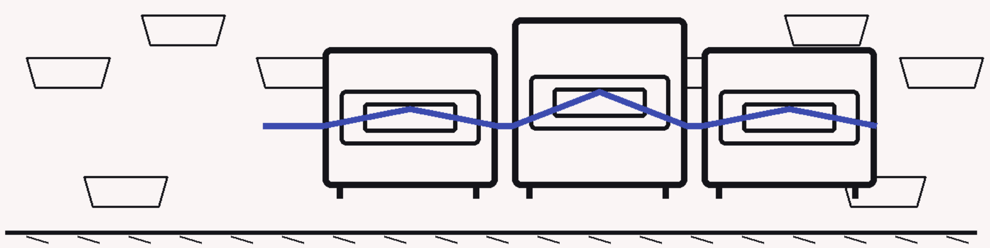
Across the locked panel, sharing anchors across guided models required 72.4% fewer DFT evaluations.

FROZEN PUBLIC ECONOMICS

72.4% fewer DFT evaluations

This bounded result applies to the locked evidence record. It is not a universal savings claim, and no additional economics are inferred from it.

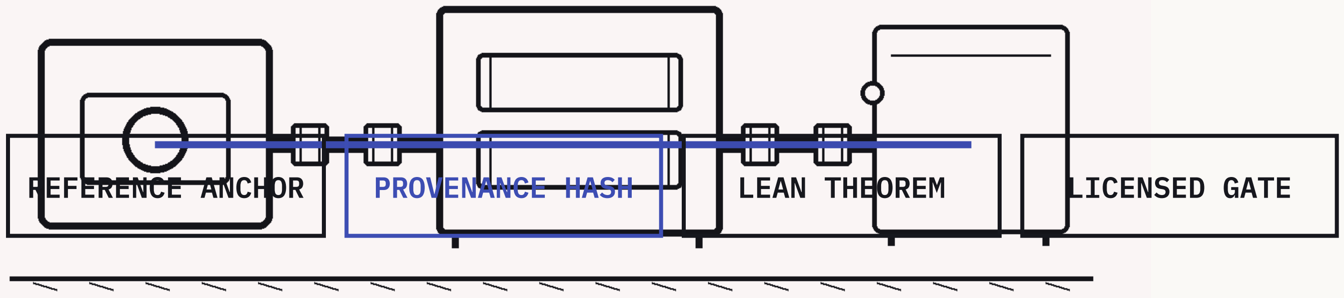
EVIDENCE-LOCKED CLAIM · NO EXTRAPOLATION



Lupine Science, “The Savings Stack,” lines 26–40; “The Union Verdict,” lines 32–35.

Physical-law theorems are non-rival: partners can contribute verified laws without disclosing proprietary chemistry, and every participant's gates sharpen.

The evidence chain ties quantitative claims to provenance-hashed data and machine-checked Lean theorems. Each failed correction and impossibility result sharpens the boundary rather than disappearing from the record.



100/100 unique still slots

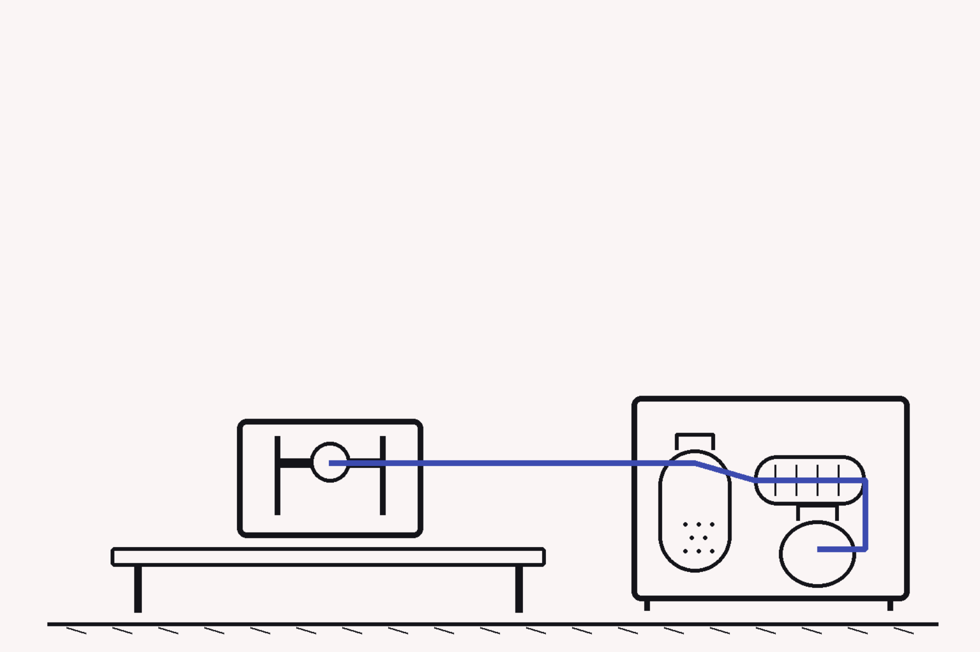
33 certified baseline stills + 67 independently certified Wave-4 replacements

Closure gate t_c2a1f8e3 · five certified films excluded from the denominator

Lupine sits between model generation and laboratory validation: model builders get external verification; laboratories get shortlists with characterized failure modes.

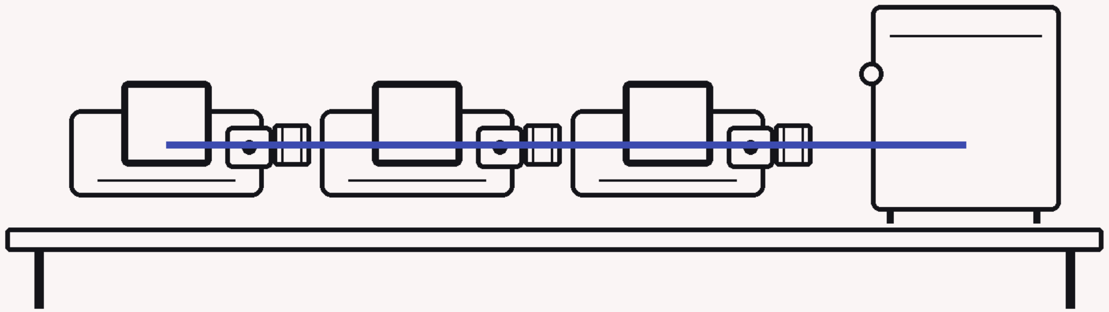
MODEL GENERATION | LUPINE TRUST LAYER | LABORATORY VALIDATION

The first market wedge is teams already running MLIP simulation in climate-critical materials. The product complements generators, databases, simulators, autonomous labs, and experimental partners rather than replacing them.



A frozen public panel, guided-model evidence, a completed execution record, and a reproducible savings analysis are inspectable now.

frozen public panel	INSPECTABLE NOW
guided-model evidence	INSPECTABLE NOW
completed execution record	INSPECTABLE NOW
reproducible savings analysis	INSPECTABLE NOW



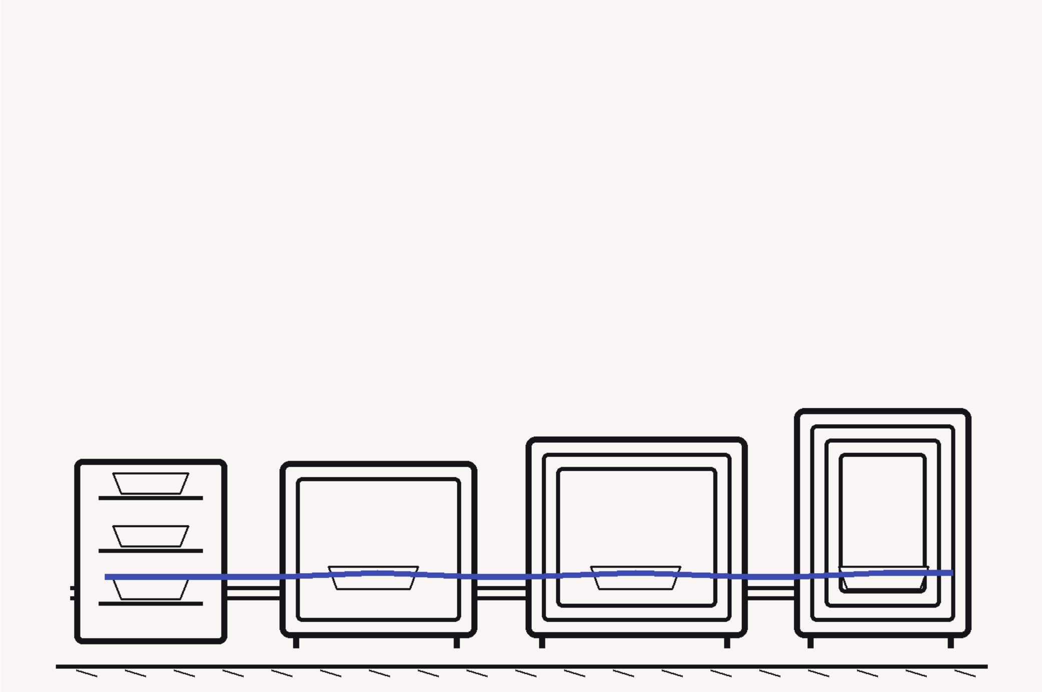
Completed evidence infrastructure — not commercial revenue or peer-reviewed validation. Z1 execution source: FOR EDITOR REVIEW — not for citation.

Lupine Science, “The Z1 Barrier Panel,” lines 40–66; “The Union Verdict,” lines 17–35 and 75–82 (FOR EDITOR REVIEW — not for citation); “The Savings Stack,” lines 26–56.

Advance by order of effort: statics → elastic response → surfaces → defects → transition states → interfaces → abstention-only domains.

STATICS	ELASTIC RESPONSE
SURFACES	DEFECTS
TRANSITION STATES	INTERFACES
ABSTENTION-ONLY DOMAINS	

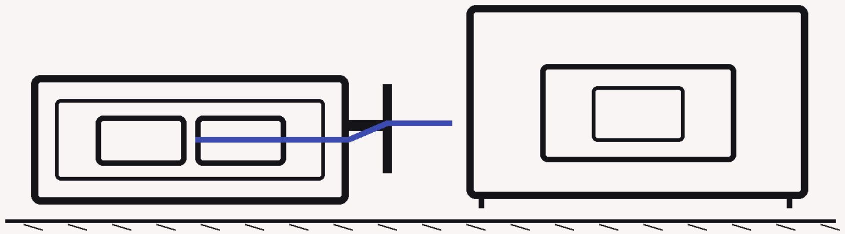
Instrument the easiest proven runtime lane; extend structure-family jurisdiction only behind explicit gates; use sparse DFT anchors where correction laws cannot act; abstain where no supported measurement path exists.



Lupine Science, “An Order of Effort,” lines 19–55.

One 30-path panel. One chemistry family. Not peer-reviewed.

- one DFT engine and functional in the executed Z1 record
- short paths make sparsity easier
- seven large paths remain deferred
- two models recorded guidance misses
- no GPAW↔VASP equivalence claim



Lupine Science, “The Savings Stack,” lines 38–42; “The Union Verdict,” lines 47–57.

Build correction-as-acceleration into the simulation loop used by every materials lab.

[OWNER DECISION] Round size
[OWNER DECISION] Instrument / terms
[OWNER DECISION] Runway and allocation

USE-OF-FUNDS FRAME

Runtime instrumentation in the proven lane.

Gated expansion across harder structure and property families.

Partner-facing evidence delivery from simulation to synthesis.

Models guide. Evidence measures.
Proof keeps the score.